

Use Cases(Phase 1)

CMPT 276-Group 15

Use Case 1: Start Game

Actor: System

Goal: To launch a new session with a new board and timer.

Preconditions: Application is running and a title screen is shown

Trigger: Player selects start game

Scenario:

1. Player clicks "Start Game" on the title screen.
2. The system creates new instance of HackingGame.
3. The system loads the board
 - Loads all room types and entity types
4. The system creates a new instance of Player at the entrance.
5. The system sets score to 0 and begins timer.
6. The system changes to the game screen.

Exceptions:

1. Board file missing or corrupted
 - System displays error message and returns to title screen
2. Required assets fail to load
 - System displays warning describing which assets failed and returns to title screen

Use Case 2: Move Virus

Actor: Player

Goal: Move the Virus one adjacent cell during 1 tick.

Preconditions: A game session is running, GameScreen is displayed, the Virus is alive.

Trigger: Player clicks a valid movement key.

Scenario:

1. Player presses a movement key.
2. The game interprets the input to move the Virus in a cardinal direction(up/down/left/right).
3. The system checks whether the adjacent cell is reachable.
4. If movement is allowed, the Virus advances into the new cell..
5. The system applies any effects associated with that cell.
6. The system updates the camera to follow the virus.

7. The system displays the updated game state

Exceptions:

1. Target cell is blocked by a barrier
 - Virus does not advance and continues from step 6.
2. Target cell contains a Sentinel
 - The game ends and the player losses, see use case "Lose Game"
3. Target cell contains a Firewall
 - Score decreases by the penalty amount and the firewall is removed from the board, see use case "Encounter Punishment"
4. Target cell contains Data Packet
 - Score increases by the reward value, the Data Packet is removed and the system checks if all required rewards are collected, see use case "Win Game"
5. Target cell contains source code
 - Score increases by the bonus reward value and the source code is removed
6. Target cell is the exit
 - System checks if all Data Packets have been collected, see use case "Win Game"

Use Case 3: Enemy Movement

Actor: System

Goal: Move Sentinels toward the Players current position

Preconditions: Game is running and not paused.

Trigger: The game processing a new tick in the main game loop..

Scenario:

1. The system calculates the shortest path from the Sentinel to the Virus each tick.
2. Each Sentinel advances one cell in the direction of the Virus.
3. If the Sentinel shares the same cell as the Virus, the game ends.
4. The system updates the game time and refreshes the display of all entities.

Exceptions:

1. Target cell contains another enemy
 - Enemy does not move and waits until the path is clear.
2. Target cell is blocked by a barrier
 - The enemy will try another direction

Use Case 4: Collect Reward

Actor: Player

Goal: Collect Data Packets and Source Codes to increase score.

Preconditions: Game is running, Reward entities are spawned, Player entity exists on the board.

Trigger: Player moves to a cell containing a Reward entity.

Scenario:

1. Player moves to a cell containing a Reward entity.
2. The Reward entity is removed from the board and retrieved by the system.
3. The system adds points to the score.
4. The system checks if all data packets have been collected.
5. If so, the game ends.

Use Case 5: Encounter Punishment

Actor: Player

Goal: To handle consequences when the players Virus encounters a Firewall

Preconditions: Firewall exists on board, player entity exists on board

Trigger: Firewall exists on player entity

Scenario:

1. Player moves to a cell with a Firewall.
2. System checks if the player position matches with the Firewall position.
3. The system applies a penalty to the players score.
4. The system checks if the score is ≥ 0 .
5. The Firewall is removed from the board.

Exceptions:

1. Score drops below 0
 - The system determines that the player lost the game and triggers the game over sequence, see use case "Lose Game"

Use Case 6: Win Game

Actor: Player

Goal: To successfully collect all Data Packets and reach the exit cell.

Preconditions: All Data Packets are collected, the Player is still alive.

Trigger: Player moves to the exit after collecting all data packets.

Scenario:

1. Player moves to the Exit cell.
2. System checks if all necessary Data Packets have been collected.

3. The system stops the game timer and records the final time.
4. The system opens the Win Screen and displays final stats.

Exceptions:

1. Not all Data Packets have been collected
 - The system prevents exiting and the system will communicate that you do not have enough Data Packets. The game will continue normally.

Use Case 7: Lose Game

Actor: System

Goal: Handle game termination when the player fails the mission.

Preconditions: Game session is running, Player is alive

Trigger: Player runs into sentinel OR score drops below 0.

Scenario:

1. The losing condition is triggered.
2. The system stops the game timer.
3. The system stops accepting input.
4. The system records the final score.
5. The system shows the Game Over Screen.
6. The system displays options to Retry, Return to Title Screen, or Exit application.

Exceptions:

1. Collision with enemy
 - Game over message specifies "Caught by Sentinel"
2. Score drops below 0
 - Game over message specifies "Virus Corrupted"

Use Case 8: Pause/Resume Game

Actor: Player

Goal: Temporarily suspend and resume game.

Preconditions: Game Screen is displayed.

Trigger: Player presses the ESC key during a game session.

Scenario:

1. Player clicks ESC during gameplay.
2. The system stops accepting player input.
3. All entities are frozen.
4. The time is paused.

5. The pause screen is brought up.
6. The pause screen displays "Game Paused" message, current score and time left, Options Menu, Resume Game button, Access Game Options, Return to Title Screen, and Exit Application.

Exceptions:

1. Player selects "Options" from menu
 - Player can modify game options, see use case "Configure Game Options"
2. Player selects "Return to Title"
 - The system confirms if the user wants to abandon the game and will return to the Title Screen.
3. Player selects "Exit Application"
 - The system confirms if the user wants to leave the application, see use case "Exit Game"

Use Case 9: Configure Game Options

Actor: Player

Goal: Customize game settings and preferences

Preconditions: Player is on Title Screen, Pause Screen, or Options Screen

Trigger: Player clicks "Options" button from the Title or Pause Screen

Scenario:

1. Player clicks "Options" button.
2. The screen displays Sound/Music volume controls, display settings, control keys, graphics quality.
3. Player clicks "apply" to save changes.
4. System updates any changes done.
5. Player clicks "Close" button.
6. System returns to parent screen.

Exceptions:

1. User selects "Cancel"
 - System discards unsaved changes, return to parent screen
2. User selects "Reset to Default"
 - Load default GameOptions values, return to parent screen

Use Case 10: Exit Game

Actor: Player

Goal: Close the game application

Preconditions: Title or Pause Screen is open

Trigger: Player clicks "Exit" from the Title or Pause Screen

Scenario:

1. The Player clicks the "Exit" button.
2. The system terminates all instances and loops.
3. Application closes.